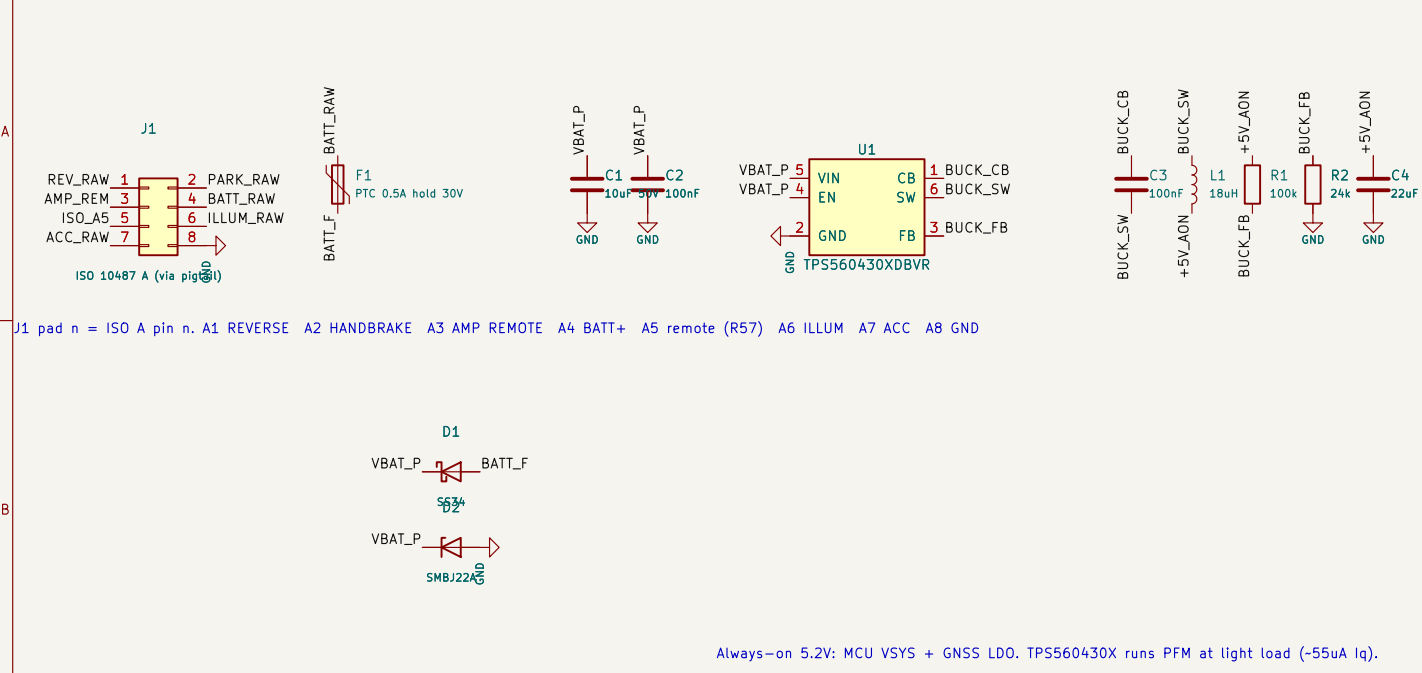


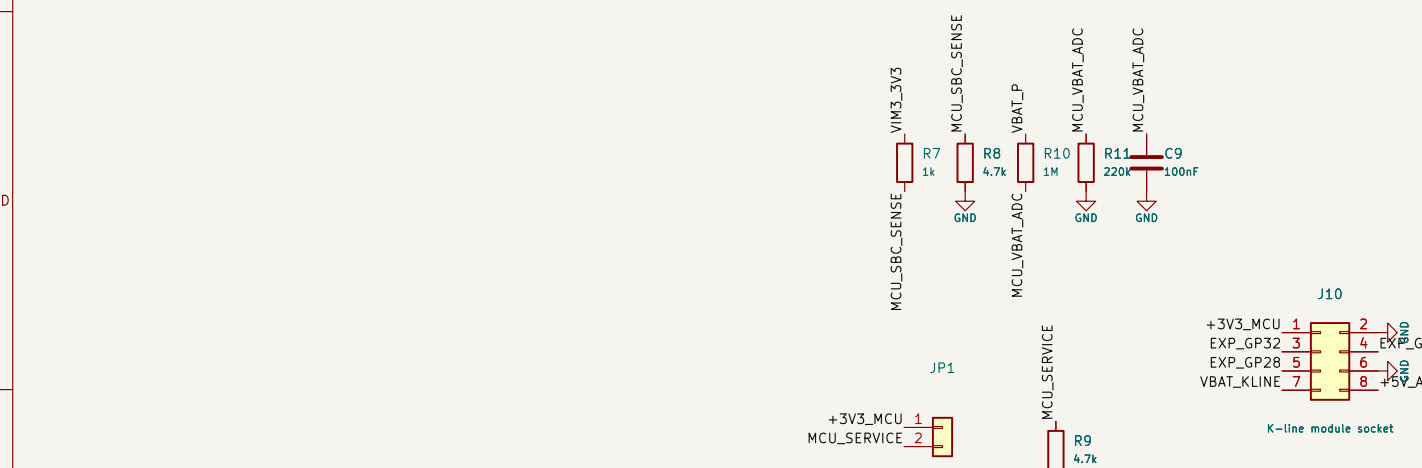
1. ISO 10487 CONNECTOR A (power) + ALWAYS-ON 5V FOR PICCO AND GNSS



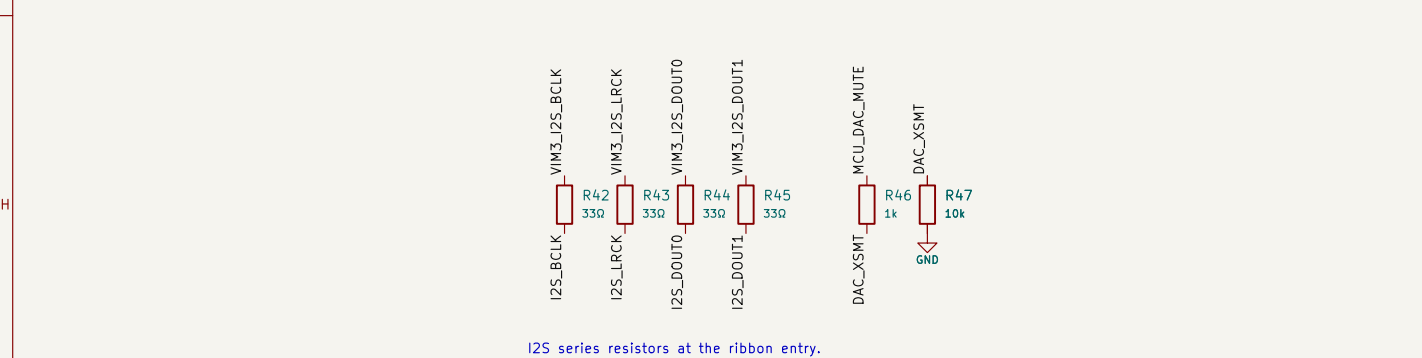
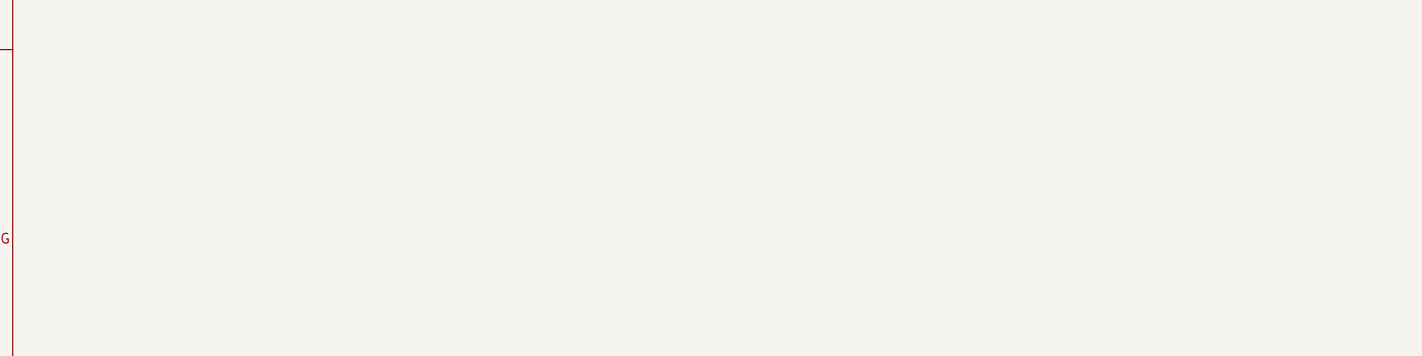
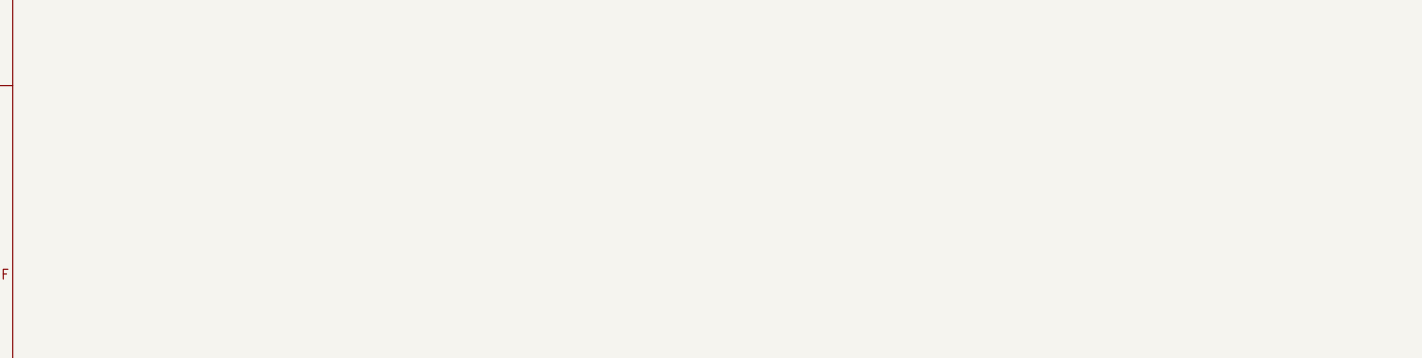
1 pin A = ISO A pin 4, A1 REVERSE, A2 VANDERBILT, A3 AMP REMOTE, A4 BATT+, A5 VEHICLE (BATT), A6 ILLUM AT ACC, A8 GND



4. MCU HELPERS (MCU core is section 13)

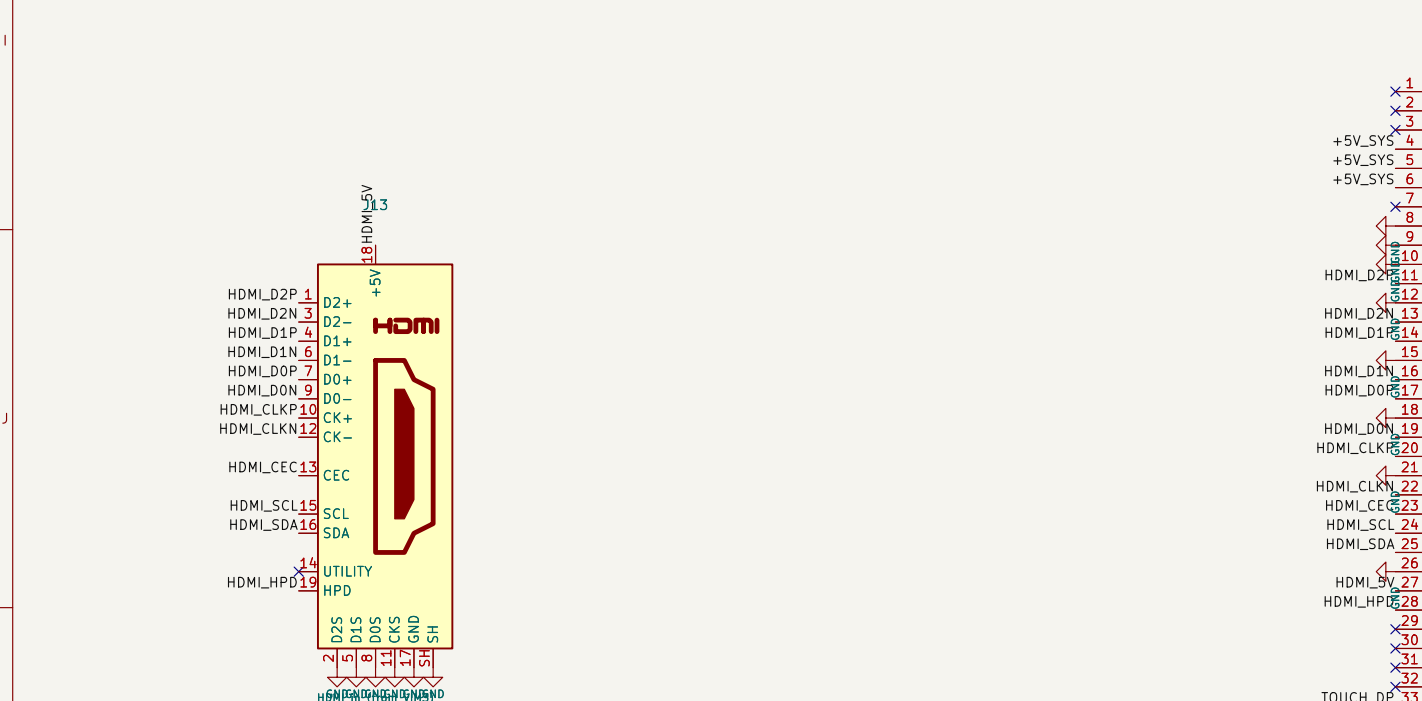


GPIO map see section 13 and DESIGN_rev03.mxd



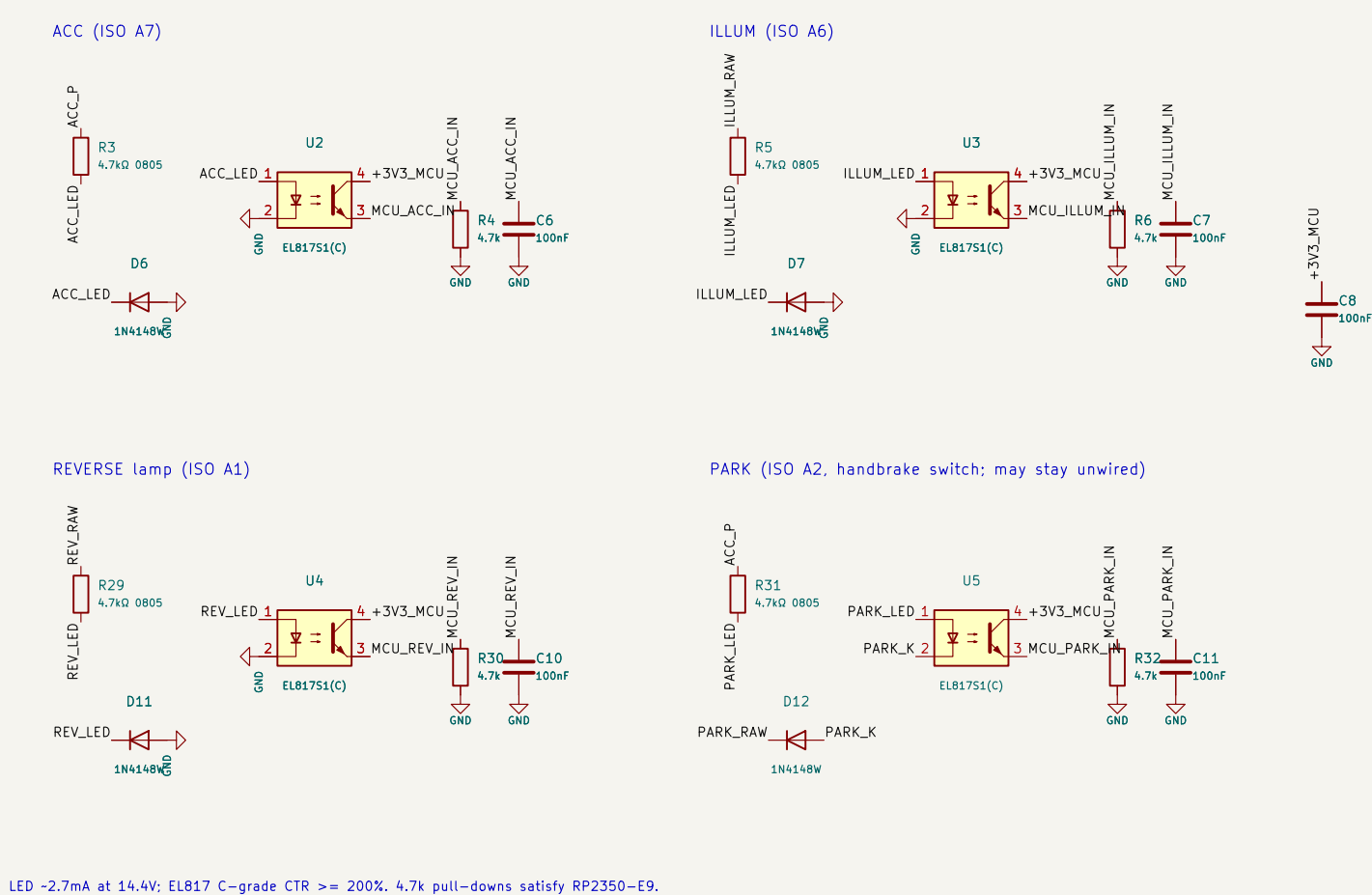
Car radio peripheral board rev 0.4 - PG F (1998), VIMS = RP2350B, 4ch line out on the audio module, NEO-M5M. Generated by genkey: use DESIGN_rev03.mxd.

5. DISPLAY: Waveshare Tin 70H-1024x600 adapter built in (HDMI in from VIMS -> 40-pin FPC to the panel)



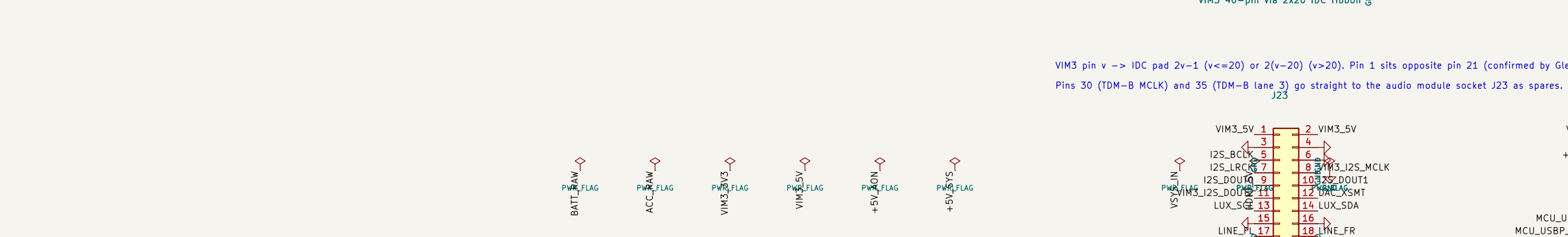
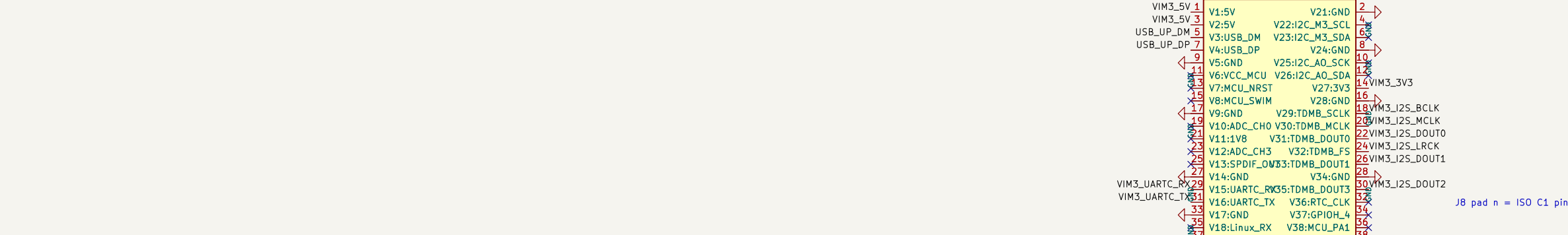
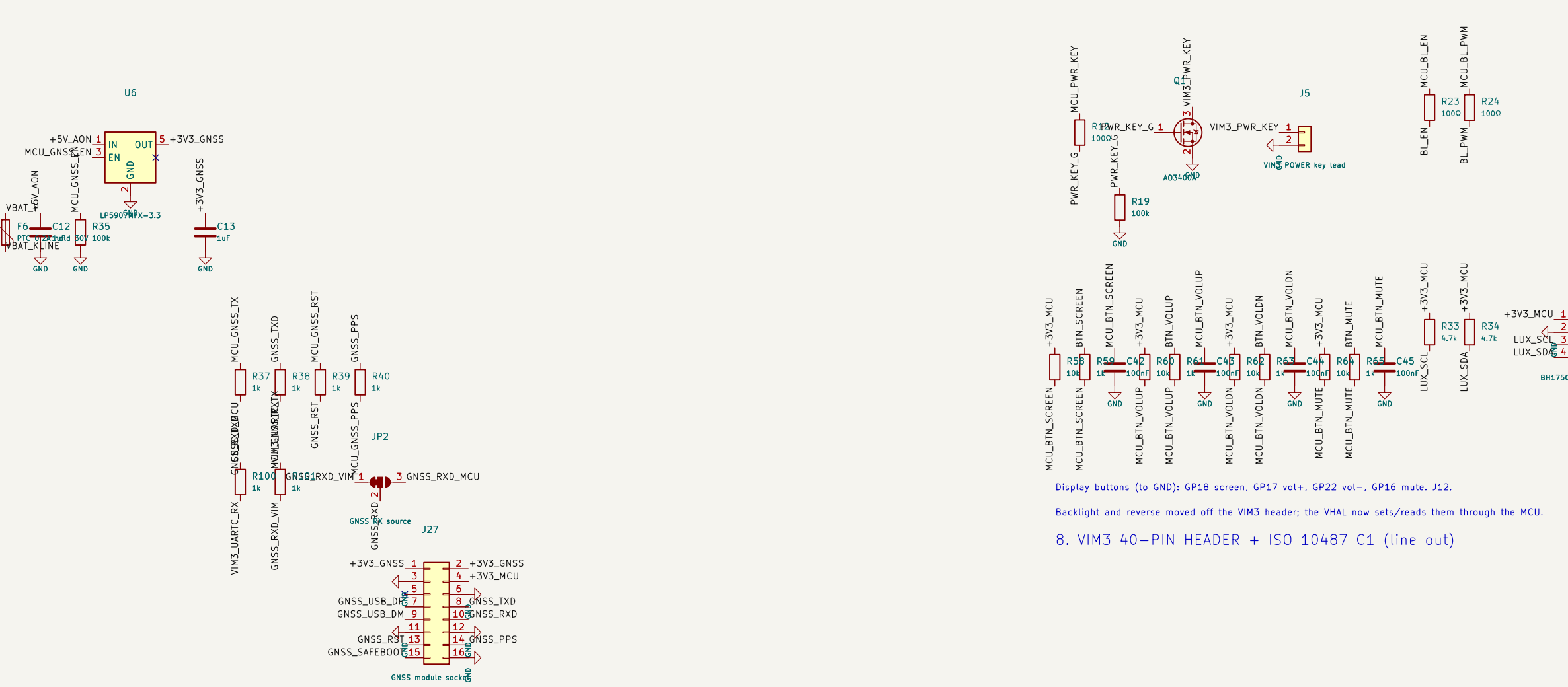
Panel 5V (0.6A) comes from the board's switched 5V rail (49/235, section 10), not from VIMS USB. Touch USB goes to the on-board hub (section 11). Backlight EN/PM now go straight to the panel 5V -> Integrated, 5.2V -> External (Panel PM is Forward), IR remote. FPC connector as in the Waveshare adapter: seen from the top, ribbon leaving the board edge pin 1 is on the RIGHT, pin 40 on the left.

2. VEHICLE INPUTS (opto-isolated, ACC, ILLUM, REVERSE: +12V = on, PARK: switch to ground = on).



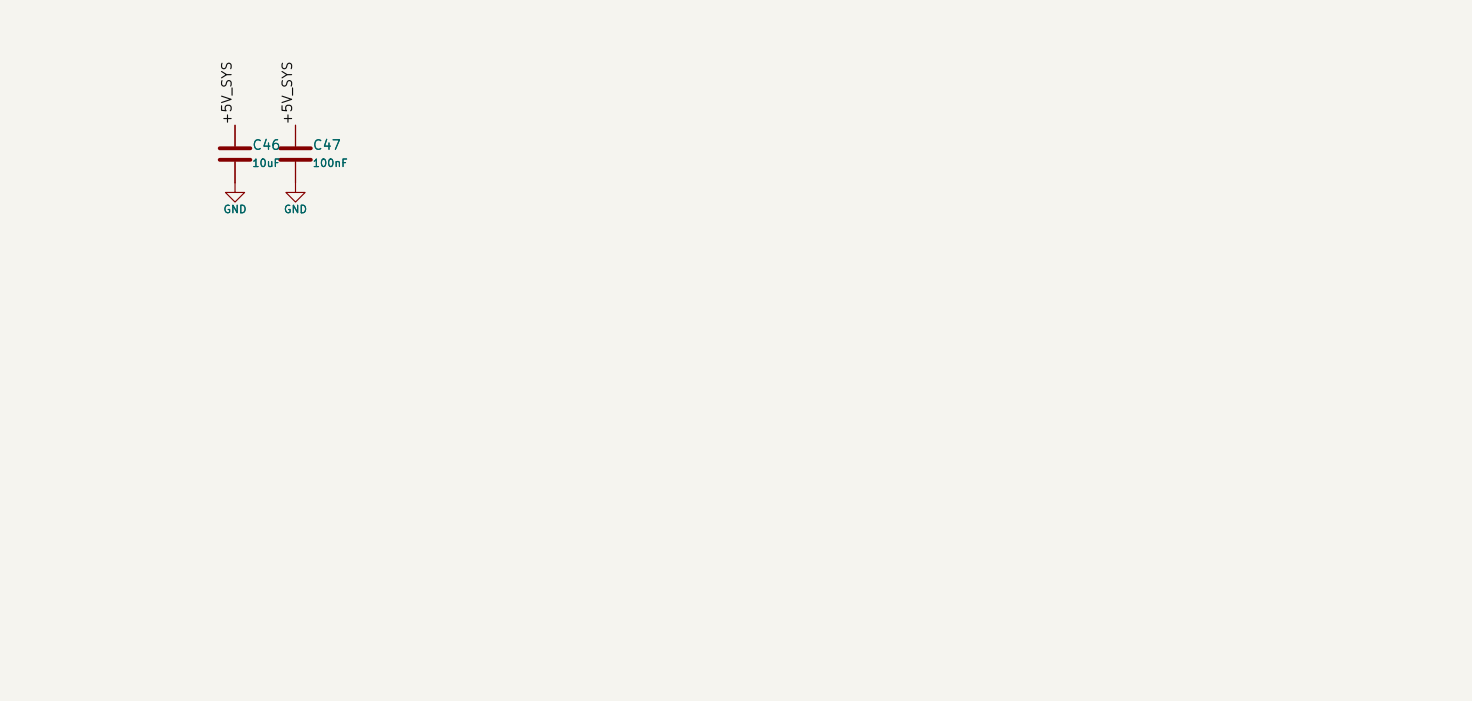
LED -2.7mA at 14.4V, SUBT C-gate CTR > 2000, 4.7k pull-down safety RP2350-15.

5. GNSS MODULE: u-blox NEO-M5N, active antenna on U.FL (dipalgal to SMA), UART + USB to the main board. Switched 3.3V supply stage VIMS3+POWER, RED6/BACKLIGHT, LIGHT SENSOR (all from the MCU)

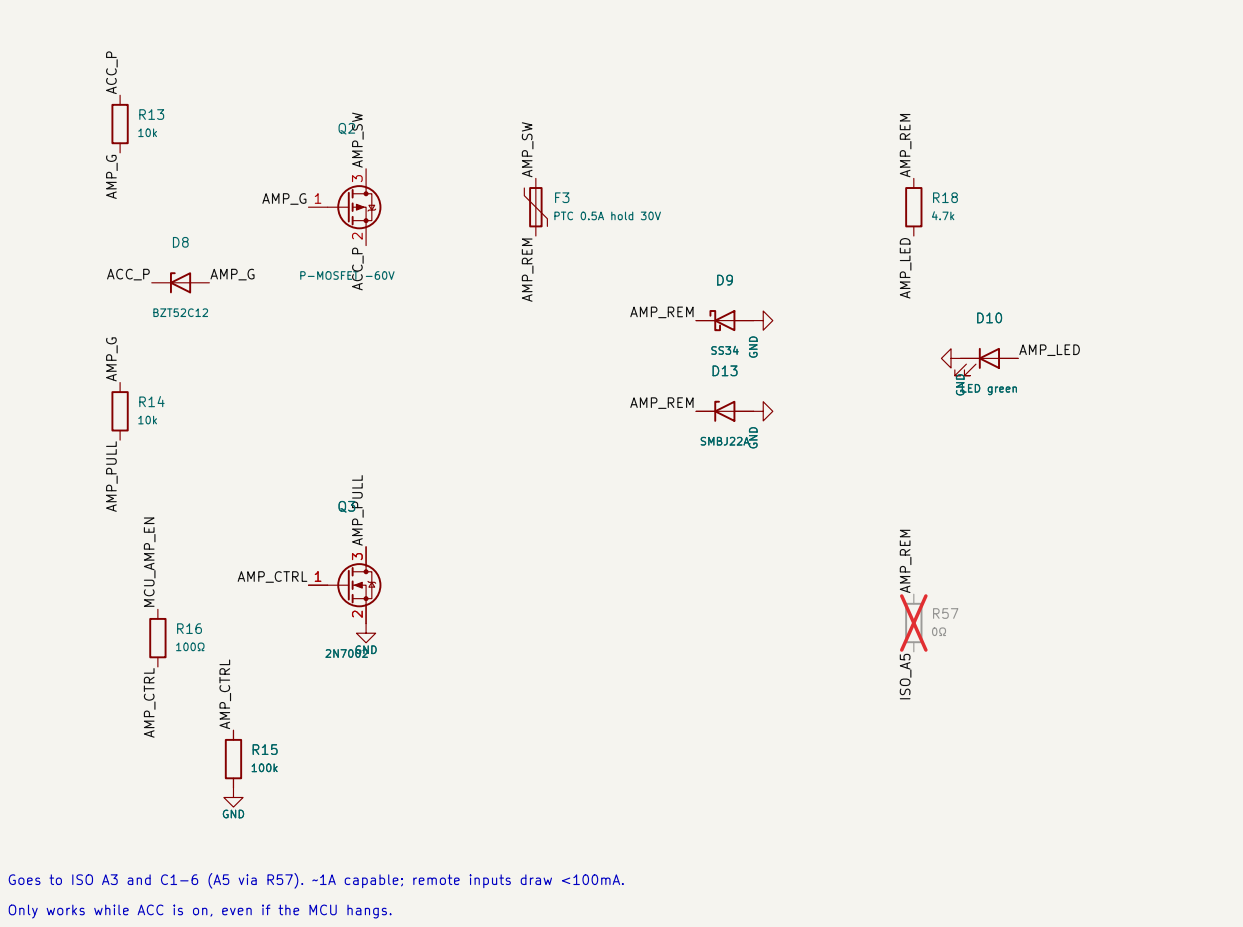


Car radio peripheral board rev 0.4 - PG F (1998), VIMS = RP2350B, 4ch line out on the audio module, NEO-M5M. Generated by genkey: use DESIGN_rev03.mxd.

5. DISPLAY: Waveshare Tin 70H-1024x600 adapter built in (HDMI in from VIMS -> 40-pin FPC to the panel)

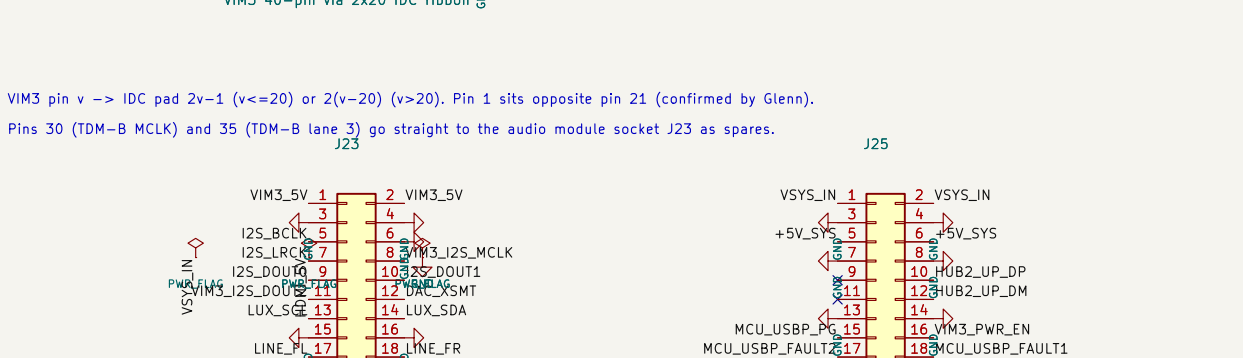
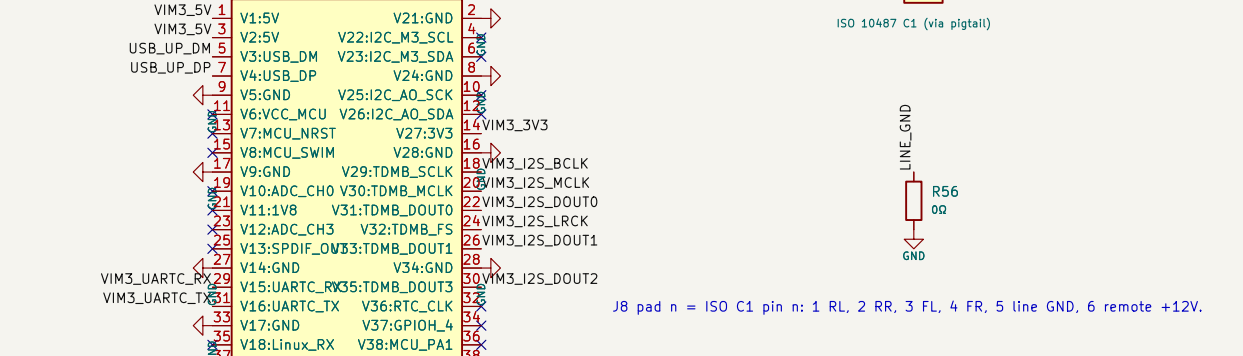
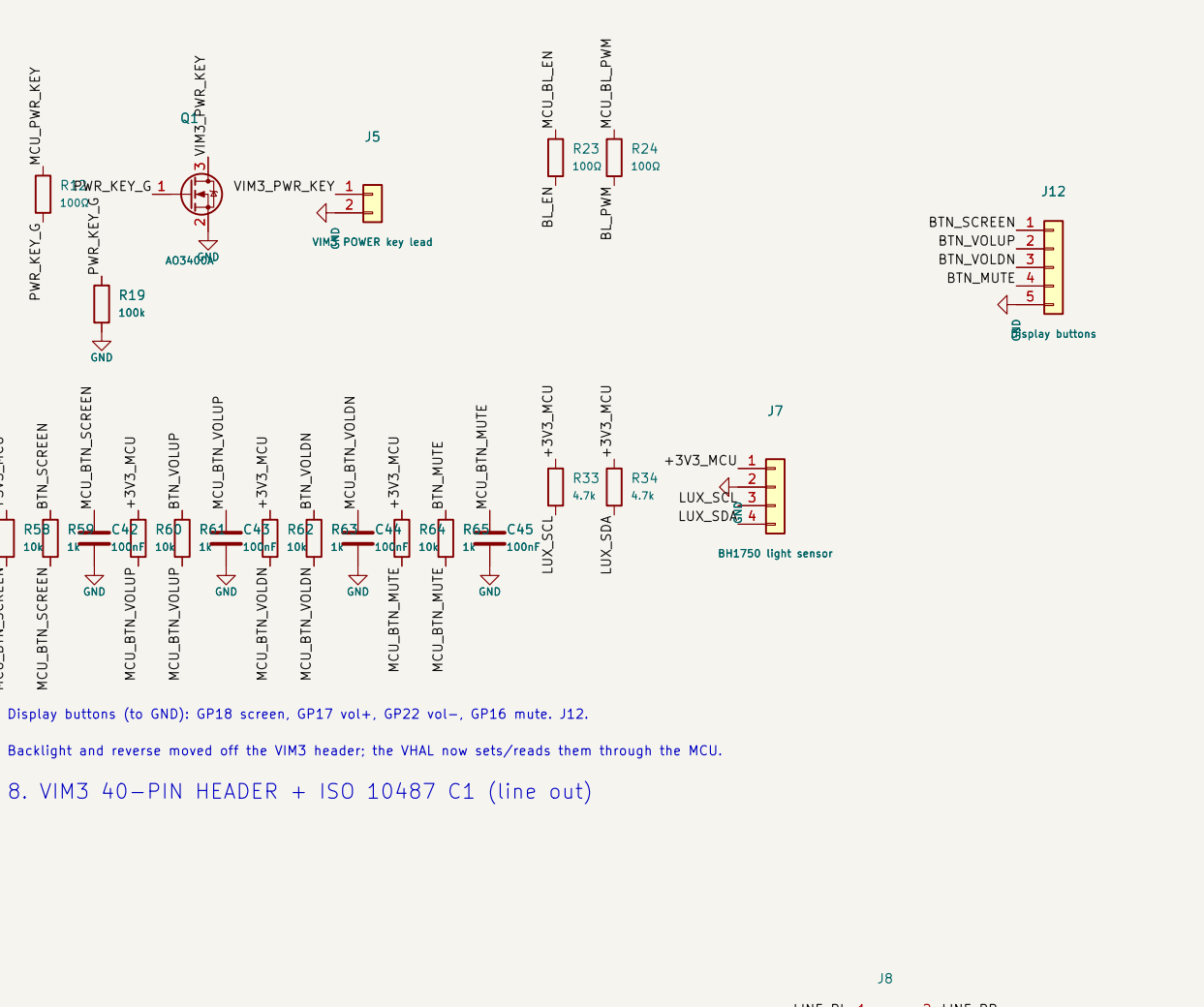


3. AMPLIFIER REMOTE (+12V from ACC, MCU GP20)



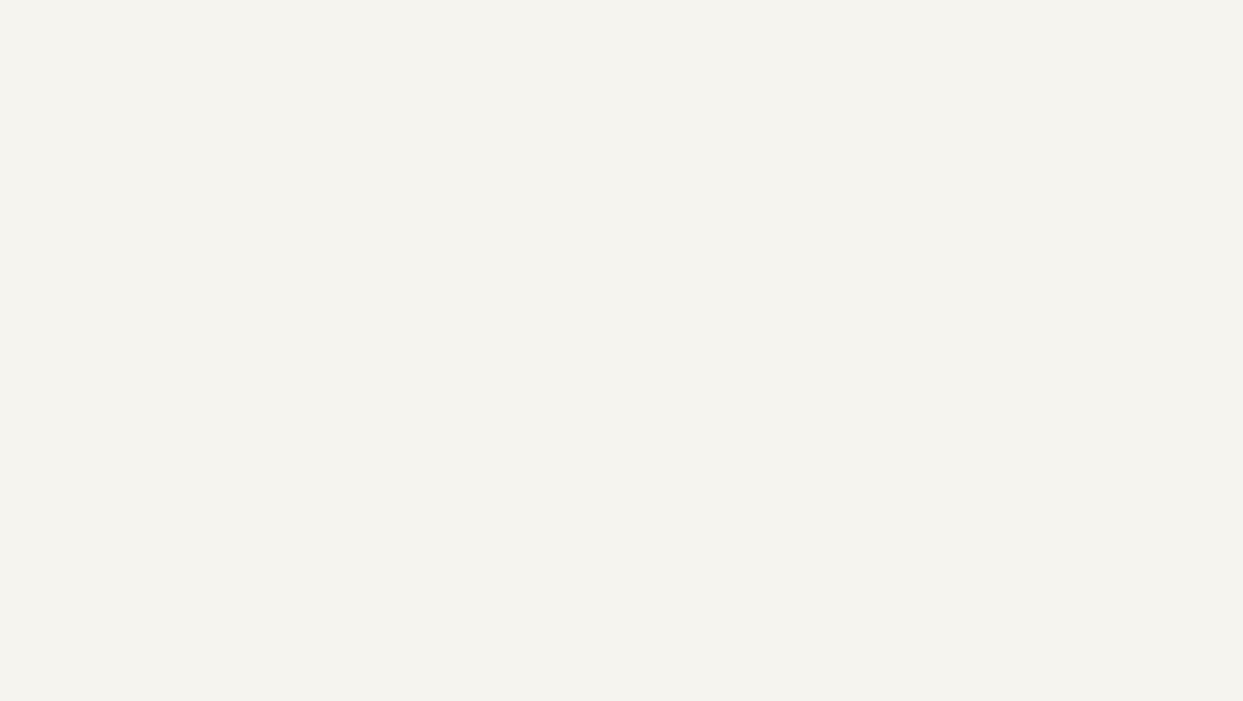
Open to ISO A3 and C4 < 40k Ohm, 4.7k pull-down safety RP2350-15.

5. GNSS MODULE: u-blox NEO-M5N, active antenna on U.FL (dipalgal to SMA), UART + USB to the main board. Switched 3.3V supply stage VIMS3+POWER, RED6/BACKLIGHT, LIGHT SENSOR (all from the MCU)

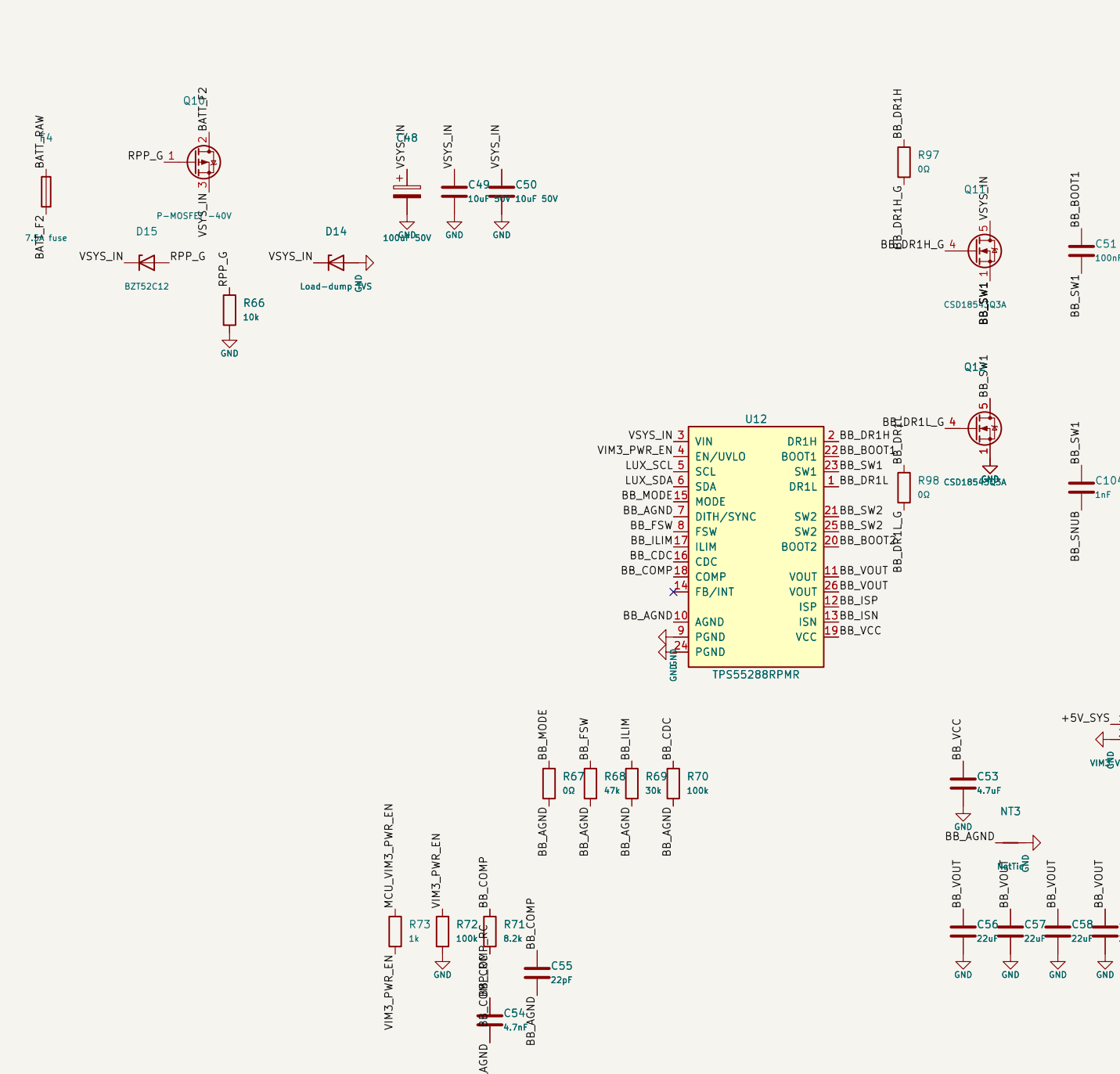


Car radio peripheral board rev 0.4 - PG F (1998), VIMS = RP2350B, 4ch line out on the audio module, NEO-M5M. Generated by genkey: use DESIGN_rev03.mxd.

5. DISPLAY: Waveshare Tin 70H-1024x600 adapter built in (HDMI in from VIMS -> 40-pin FPC to the panel)



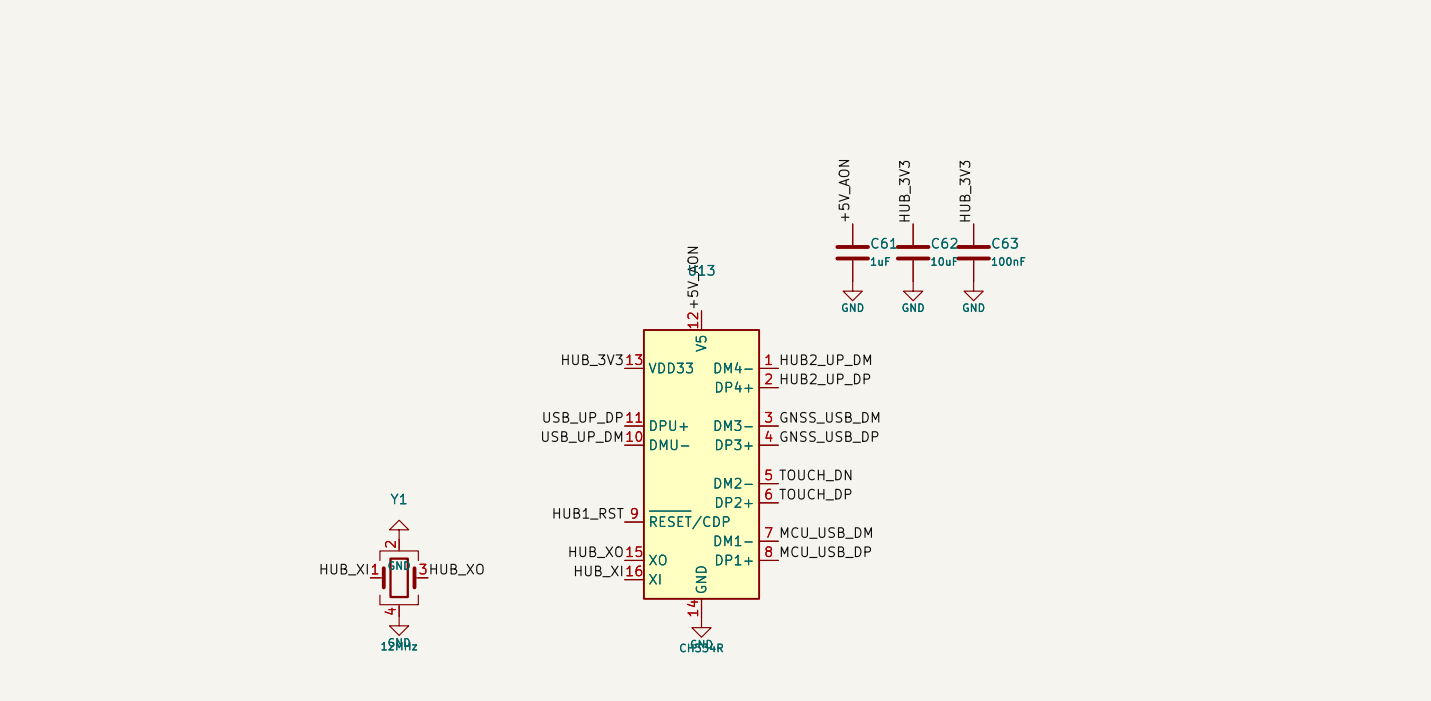
10. 5V/5A SYSTEM SUPPLY FOR VIMS3 + SCREEN + HUB (TPS55288 buck-boost; rides through cranking and overvoltage)



Input 3V-15V (loads through cranking) output 5.0V up to 5A, MCU GP22 < 40k Ohm, 4.7k pull-down safety RP2350-15.

VN is permanent +12V so the VIMS can finish its bootload shutdown after ACC drops.

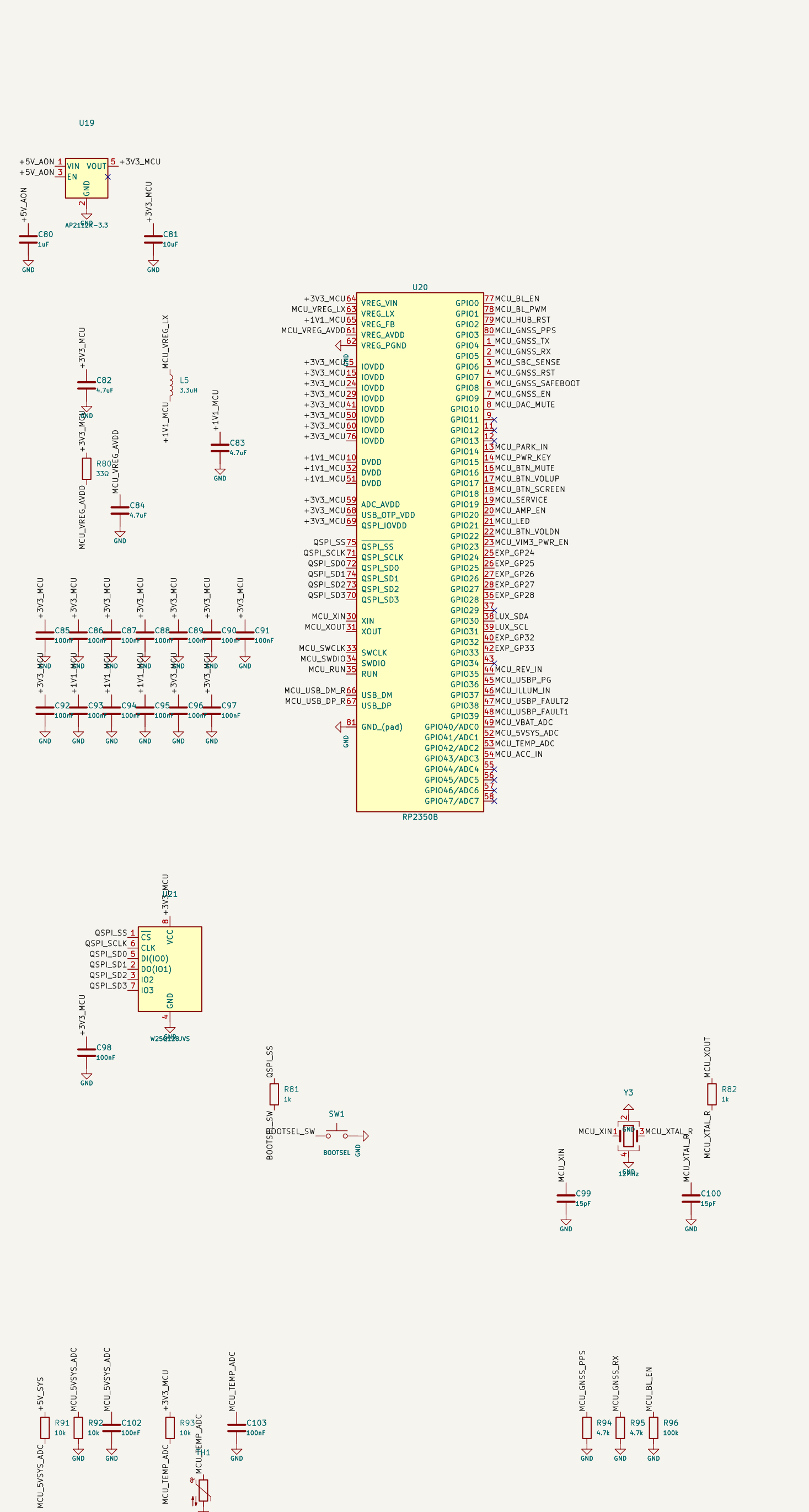
11. USB HUB: VIMS header USB (pins 3/4) -> MCU, screen touch, spare port



Hub 1: none from the always-on +5V, 200mA no a break MCU overcurrent (that flash over USB). One pin to the VIMS, no USB cables. Part 1 MCU, 2 touch, 3 GNSS module USB, 4 second hub (external ports = RP2350, Route 2 to 3 -> as 80 pin pins).

VIMS header pins 3/4 are a working USB host pins (confirmed by Genkey, 2024-09-29).

13. MCU: RP2350B (QFN-80) + 16MB QSPI flash + 12MHz crystal + 3.3V LDO, per the Raspberry Pi Pico 2 reference design



rev 0.4: GPIO map re-assigned for routing, see the DESIGN pin table, ACC, GP20A/PS, QC, IRB pulled up.

GP20A-47 are asserted (BCC-asserted signal).

RP2350-49 check (every MCU pin) outputs GP22/23/24/25 4.7k down; service jumper 4.7k down; buttons, USB Host/PS, QC, IRB pulled up.

QC11 PPS/EX 4.7k down (GP24/25) outputs have default off-state resistors; ACC pins run with the digital input disabled; unused and expansion pins see DESIGN_rev03.mxd.